

THE UNIVERSITY OF TEXAS AT AUSTIN
Department of Chemical Engineering

ChE 372 (Unique No 15410)
Chemical Reactor Analysis and Design
Spring 2024

Meeting: *Tuesdays and Thursdays 9:30 to 11:00 am Burdine 216*

Recitations: *Wednesdays 12:00 to 1:00 pm. Burdine 216*

Instructor: Prof. Nicholas A. Peppas
Office: BME 3.503C
peppas@che.utexas.edu;
Tel: 512-633-6740.
Office Hours:
Tuesdays and Thursdays in Peppas room at 12:30 to 2:00 pm

Teach. Assistant: Fabiola Chapa Villarreal
fabiola.chapavillarreal@austin.utexas.edu
Office Hours:
Mondays and Wednesdays at 4:00 to 5:00 pm in BME room 5.300

Grader: Zoe Min
zoemin19@gmail.com

Textbook: C.G. Hill Jr. and T.W. Root, "Chemical Engineering Kinetics and Reactor Design," second edition, Wiley, New York, 2014.

Grades: Three exams (3 x 100) = 300 points
Homework Sets (6) = 100 points
Final exam = 100 points

Students will be allowed to **drop the lowest of four exams** grades.
Then the total points will be 300+100 = 400 points

Course ChE 372 Descriptor

Course Objectives:

You will learn how to come up with the kinetics of simple and complex reactions. You will learn how to derive and apply the energy and material balances that are required to design isothermal and non-isothermal batch, plug flow and continuous stirred tank reactors. You will learn how to approach and solve volume dependent and multiple reaction problems.

The course presents also the fundamentals of reaction stoichiometry, reaction analysis, and simple kinetic analysis of homogeneous and heterogeneously catalyzed reactions. The course addresses mixing and covers simple, one-parameter non-ideal reactor models.

Catalytic reaction analysis and the dynamic behavior of reactors are two subjects of major importance to this course.

Teaching Approach:

Chemical reaction analysis and reactor design is unique to chemical engineering. There are relatively few concepts and design equations that are needed to describe most situations at the undergraduate level. However, there are many different reactors and reactor situations. This course starts with a complete analysis of kinetics and then with a rigorous mathematical development of the defining design equations and then this set of equations is used to solve problems in the different reactors and reactor situations. The course strives to show how the simple and complex problems are solved with the same set of equations and how the concepts learned on one system are directly applicable to other systems.

During this semester you will need to solve many of the complex and some of the simple problems using MATLAB. Through the homework exercises you will apply the concepts presented in the lectures and in the text, and should recognize that problems you will encounter in the future may be more time consuming to solve, but that they will not be any more difficult conceptually.

Knowledge, Abilities, and Skills Students Should Have Entering this Course

1. This course has the following prerequisites: ChE 322, 348 and 354 with a grade of at least C.
2. A comprehensive knowledge of transport phenomena is essential. This course relies on controlled volume balances and the constituent equations to develop the material and energy balances for the various reactors, and to develop the balances in heterogeneous catalysts that account for diffusion with reaction. Transport phenomena concepts are used to appreciate the consequences of turbulent flow on velocity, temperature and concentration profiles that form the basis of the one-dimensional models. (ChE 353)
3. Knowledge of heat transfer processes is needed for non-isothermal reactors. (ChE 353)

ChE 354)

4. Understanding of reaction equilibrium and heat of reaction is required. (ChE 322)
5. The students need to be able to solve linear differential equations (M427K)
6. The students need to know matrix multiplication (ChE 348)
7. The students need to know how to solve ODEs using explicit numerical methods and to have a working knowledge of how these methods work (ChE 348)
8. The students need to know how to solve sets of linear and nonlinear algebraic equations using numerical methods (ChE 348).
9. The students need an appreciation of numerical optimization procedures (ChE 348)
10. The students need to have a working knowledge of MATLAB numerical software that operates on the Department Learning Resource Center computers (ChE 210 and ChE 348).

Knowledge, Abilities, and Skills Students Should Gain from this Course

1. Ability to derive and apply the energy and material balances that are required to design isothermal and non-isothermal batch, plug flow, fixed-bed and continuous stirred tank reactors.
2. Ability to solve problems of variable volume and multiple independent reactions.
3. Ability to solve problems of mass transfer with reaction in heterogeneous catalysts.
4. Ability to solve problems incorporating simple, one-parameter nonideal reactor modeling.
5. Knowledge of chemical reaction analysis concepts and chemical kinetics concepts.

Chemical Engineering Program Outcomes Achieved

1. For ChE 372, outcomes of the course will be that students have
 - a. the “ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics”, and
 - b. the “ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.” (These are ABET Student Outcomes 1 and 2).
2. The AIChE defines a complex engineering problem as having one or more of the following characteristics: “involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.”

ABET Outcome Evaluations

Course Course: ChE 372

Course Name: Chemical Reactor Analysis and Design

Semester: Spring 2022

ABET criterion being evaluated: **Student Outcome 1** - an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

In selected assignments homework and quizzes: Professor Peppas will prepare a brief paragraph describing how the exam question or course assignment measures the above ABET Criterion, specifically focusing on why the problem is complex, where complexity is defined as having one or more of the following characteristics: “involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.”

ABET criterion being evaluated: **Student Outcome 2** - an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

In selected assignments and homework: Professor Peppas will prepare a brief paragraph describing how the exam question or course assignment measures the above ABET criteria. AIChE specifies that “the intent is to include all of these issues in the same design – however, it is not necessary that they are all relevant.”

Course outline:

The course outline is attached. You are required to read the assigned text material as listed in the document entitled “Tentative Schedule” and all the handouts provided, ***even if they are not covered to a great extent by Professor Peppas***. In no way should Prof. Peppas’ lectures be considered as a simple repetition of what is in the book.

Attendance:

You are expected to attend the course lectures and recitations although no attendance will be taken. Attendance of the recitation sessions is highly recommended as various problems will be solved.

Homework and Exams:

1. All required homework, case studies and exam dates are clearly indicated in the first page of this handout.
2. ***There will be no make-up exam or change of scheduled exam dates.***
3. Material asked in an exam is not a simple repetition of what was solved in the recitation hours or discussed/solved in class. ***The exams of this course test you on***

your ability to take what has been discussed in class and apply it to other similar problems or to different situations whose mathematical basis has been already addressed in class. When an exam problem requests exact numerical values, you are expected to carry out all calculations and arrive to a reasonable numerical answer, otherwise you will receive no more than 25% of the maximum grade. In ChE 372 exam, it is the Professor's practice to treat you as a young engineer. ***Thus, you will not be given material that is considered standard knowledge of a young engineer, such as the molecular weight of a simple compound, the value of the universal gas constant, how to take a simple derivative, etc.***

4. ***If you do not turn in a homework solved, you will receive a zero.***
5. All homework will be graded by the grader. Regrading of the homework solution will be considered only if ***a written request is submitted to Prof. Peppas*** within 48 hours from the time the specific homework has been returned to you. Without a written request explaining your disagreement with the numerical grade given, no regarding will be done.
6. All examinations will be graded by N.A. Peppas. Regrading of the examinations will be considered only if submitted in writing within 48 hours from the time the specific examination has been returned to you. Without a written request explaining the disagreement with the numerical grade given, no regarding will be done. No request will be considered or discussed if presented just verbally, in passing, or long past the examination period.
7. Participation in class discussion is highly encouraged.

Grading Policy:

Students will be allowed to drop the lowest of four exams grades.

Then the total points for the course will be 300 points for exams and 100 points for homework = 400 points

Grading is based on placing all grades of all the students on a curve and assigning grades based on comparative performance. This professor is strict with his grading system. For required undergraduate chemical engineering courses, his typical evaluation in the past has been 30-40% A or A-, 40 % B+, B or B-, and 20% C+, C, C-, D or F.

Dropping Class:

Familiarize yourselves with the CSE practices for dropping a class late in the semester. ***Please note that Professor Peppas will not allow dropping with a passing grade late in the semester if the student has a performance equivalent to a D or F at the time of class drop. Consider this policy very carefully before you drop the class.***

Office Hours:

Students are urged to use the announced office hours to discuss their questions and problems with the professor and TAs. ***Students are expected to use the office hours of Professor Peppas to ask questions about the course, individual lectures, etc.***

Questions about the way a particular homework problem should be approached should be addressed to the TA during their office hours. Do not expect that the TA will solve a problem for you!

Questions will be answered only if you have tried the problem and you have legitimate concerns. The TA will not solve the problem for you.

Other Information:

The University of Texas at Austin provides, upon request, appropriate academic adjustments for qualified students with disabilities

Scholastic Dishonesty: The official University policies can be found at <http://www.deanofstudents.utexas.edu/sjs/academicintegrity2.html>. We urge you to avoid collaboration with other students either in homework or in the Case Study.

Emergency Preparedness and Emergency Plan Instructions:

Please take note of all emergency exits, exit procedures, and University Emergency Communications Policies for the lecture and recitation buildings. Students requiring assistance in evacuation should inform the Professor of the course in writing of their needs during the first week of class so that this information is properly transmitted to the Fire Prevention Services office.

UNIVERSITY OF TEXAS
ChE 372 (Unique No 15410)
Spring 2024

TENTATIVE SCHEDULE OF LECTURES

Week	Date	Subject	Reading (pages)	Homework Due
1	1/16	Stoichiometry, Review of Thermodynamics, Equilibrium	1-15	-
	1/18	Basic Concepts in Chemical Kinetics: Determination of Reaction Rate Expressions	19-26	-
2	1/23	Determination of Reaction Rate Expressions and Reaction Rate Expressions	26-35	-
	1/25	Interpretation of Reaction Rate Data	35-53	Set No.1
3	1/30	Molecular Interpretation of Kinetic Phenomena Chain Reactions, Bodenstein Approximation	72-83 83-93	-
	2/1	Molecular Theories	93-103	-
4	2/6	Reversible Reactions	117-125	-
	2/8	Parallel Reactions, Irreversible Reactions. Complex Reactions	125-133, 133-142	Set No. 2
5	2/13	<i>Overview of Material of weeks 1 through 4</i>	<i>1-142</i>	-
	2/15	EXAM No. 1		Exam No. 1
6	2/20	Heterogeneous Catalysis, Absorption Isotherms.	152-160	-
	2/22	Reaction Rates - Catalyst Preparation	160-178	-
7	2/27	Liquid Phase Reactions. Enzymatic Reactions	188-205	-
	2/29	Introduction to Reactor Design	216-228	Set No. 3

Week	Date	Subject	Reading (pages)	Homework Due
8	3/5	Material and Energy Balances. Batch Reactors	216-234	-
	3/7	Tubular Reactors, CSTR Reactors	235-241	Set No. 4
9	3/19	CSTR in Series. Recycle Reactors	235-258	-
	3/21	Selectivity and Optimization	273-283	-
10	3/26	Parallel Reactions. Consecutive Reactions	283-289	Set No. 5
	3/28	EXAM No. 2	-	Exam No. 2
11	4/2	Autocatalytic Reactions. Temperature Effects	290-294, 305-307	-
	4/4	Ideal Batch Reactors	307-312	-
12	4/9	Ideal CSTR, Tubular Reactors, Stability	312-327	-
	4/11	Residence Time Distribution Function	337-343	Set No. 6
-				
13	4/16	Non-ideal Flow Situations	343-358	-
	4/19	Design of Heterogeneous Catalytic Reactions Mass Transfer in Porous Catalysts	371-376 376-406	-
14	4/23	EXAM No. 3	-	Exam No. 3
	4/25	Packed Beds. Reactor Design	371-406, 406-438	-
Finals Week	- TBA	- EXAM No. 4 (FINAL EXAM)		Exam No. 4